Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study

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***Abstract*—Large Language Models are increasingly used for automated code review, and multi-agent frameworks propose dividing the work among specialized agents. It remains unclear whether the *structure* of that collaboration—rather than the agents' mere presence, or the extra compute they consume—improves review quality; only reviews whose errors are *decorrelated* can be combined into a better one. Under a pre-registered, compute-matched design, we compare a four-rung experimental *ladder* that incrementally adds sampling, role specialization, and peer communication—an Agentless single-LLM baseline, a compute-matched Generalists-3 control, a Hierarchical manager-plus-specialists design, and a Decentralized Peer Consensus design—holding the model, PR snapshot, prompts, and evaluation pipeline fixed, across three datasets: injected-defect ground truth, human-reviewed PRs, and a live industrial repository. Three findings emerge. *Communication is not the lever:* role specialization yields no improvement over compute-matched sampling (p=0.73), and no multi-agent workflow exceeds the single-pass baseline's F1, buying recall only at 3–9× the call budget. *Verification is:* findings that independent model families agree on match injected ground truth 89% of the time versus 54% for one model repeating itself—error decorrelation paying on precision, not coverage. *Grounding bounds both:* unioning every configuration we ran plateaus at a ≈0.83 recall ceiling whose residue is mechanical conventions a linter recovers and a functional hardcore that execution, not more reading, begins to reach; and on unlabeled industrial code the verification relationship reaches its ecological boundary, because LLM correctness judges themselves disagree (κ=0.03).**

***Index Terms*—Large Language Models, Multi-Agent Systems, Automated Code Review, Evaluation, LLM-as-a-Judge.**

I. Introduction

Code review gates most software changes, and LLMs now plausibly automate parts of it. A prominent design direction decomposes the task among specialized agents that communicate, as in ChatDev [1] and MetaGPT [2]. Most evaluations, however, compare a *whole* multi-agent system against a single model, conflating additional computation with workflow structure—how many agents there are versus how they are organized. We ask:

> **RQ:** Holding the underlying model, review target, and evaluation procedure fixed, what incremental effects do additional sampling, role specialization, and inter-agent communication have on the effectiveness and efficiency of automated code review?

Answering this needs a control most studies omit: if a three-specialist team beats a single reviewer, the cause may be three samples rather than three *roles*. We therefore add a compute-matched **Generalists-3** arm—the same prompt sampled three times and merged—so specialization is tested against equal compute (Agentless remains a strong single-pass baseline [3]). This exposes the paper's organizing idea, *error decorrelation*: additional reviewers improve quality only insofar as their mistakes are independent, so the question throughout is whether a given form of structure decorrelates a reviewer's errors or merely repeats them.

Because a benchmark result is only as trustworthy as its evaluation, we evaluate on three complementary datasets—injected-defect ground truth, human-reviewer agreement, and a live industrial repository—and pre-registered the hypotheses, sample sizes, and analysis before collection. Our contributions are:

- A pre-registered adjacent-contrast evaluation (2,415 reviews) isolating sampling, specialization, and communication.
- Evidence that homogeneous multi-agent workflows improve recall but not F1, while agreement between independent model families is a precision signal.
- Evidence that grounding—not additional communication—is the remaining bottleneck: a ≈0.83 recall ceiling whose residue is lint-targetable conventions and a repo-integrated functional core, bounded by an industrial case study where the verification signal cannot be validated without an oracle (κ=0.03).

Scope and thread. The registered study isolates the workflow axis and its answer is largely null; the exploratory extensions (Sections IV-D–IV-F) trace *why*, reducing every movable lever to one axis—how *decorrelated* the reviews are—which pays on precision, not coverage. We mark confirmatory and exploratory claims explicitly throughout.

II. Related Work

LLM-based software engineering. SWE-bench established that realistic GitHub tasks are hard for language models [4], and Agentless shows a simple non-agent pipeline is competitive [3]—the natural baseline for asking whether multi-agent communication adds value.

Multi-agent software engineering. ChatDev [1] and MetaGPT [2] organize development as communication among role agents; surveys find the area promising but under-evaluated [5], and recent work catalogs *why* such systems fail—coordination overhead, cascading errors, weak verification [6]. Concurrent 2026 results agree: agent teams can hold

TABLE I
EVALUATION DATASETS.

	PRs	Reviews	Correctness signal
E1 Qodo	99	1,188	injected ground truth (P/R/F1)
E2 SWE-PRBench	50	600	human-reviewer agreement
E3 industrial	30	627	corroboration proxies (no oracle)

experts back [7], and mixing models does not beat the best single model [8].

Verification and LLM judges. LLM judges scale evaluation [9], verifier-based selection converts extra samples into quality [10], and diverse judge panels can replace a single judge [11], though judges show self-preference [12] and prompt-form sensitivity [13].

Across this literature the assumption is that better coordination or specialization yields better review, yet none isolates compute from communication or tests error *decorrelation* directly—which is what our compute-matched adjacent-contrast design does, alongside a non-circular objective/subjective judge split and a powered test of the correlated-error conjecture [14].

III. Study Design

A. The four-rung ladder

Each rung adds exactly one capability over its predecessor (Fig. 1): (1) **Agentless**—one LLM reviews the PR in a single pass; (2) **Generalists-3**—the same generalist prompt sampled three times and merged, with no roles and no communication (the compute-matched control that isolates sampling from specialization); (3) **Hierarchical**—a *deterministic* coordinator routes the diff to frontend, backend, and database specialists, whose findings a *deterministic* synthesizer merges; (4) **Consensus**—the same specialists review independently, then exchange findings and vote.

Hierarchical routing and synthesis are deterministic code, so its three LLM calls are exactly its three specialists; only Consensus adds inference stages (nine calls). The model (Claude Haiku 4.5, with a Sonnet 4.5 robustness arm in Sections IV-A and IV-E), prompts, per-PR snapshot, and evaluation pipeline are fixed; only sampling budget, roles, and message flow vary, so we read *adjacent* contrasts rather than treating topology as a single isolated variable. Each (PR, arm) value is the mean of three runs; run variance is negligible (within-PR SD of semantic F1 $\approx$0.01–0.05) and the headline ordering holds in each run individually.

B. Datasets and metrics

Table I summarizes the three datasets. E3's 627 reviews are the four-arm ladder plus single-pass cross-family arms (Haiku 4.5/Kimi K2.5/GLM 5, three runs each; GLM usable on 29/30 PRs).

Findings are matched both strictly (file+line) and *semantically* via an independent LLM judge's binary same-issue decision (threshold-insensitive across 0.5–0.9); malformed findings are dropped and logged, not failed, removing a survivorship bias against the single-call arm. Operational metrics (latency, tokens, calls, messages) are collected identically. Lacking an oracle, E3 substitutes corroboration proxies on the cross-*family* axis (the only one that carries signal, Section IV-B): *depth* clusters independent findings with an LLM pair judge (Nova Pro, τ=0.7) and counts the model families agreeing; *proxy precision* is a separate judge's genuine-rate, and *proxy recall* is leave-one-out cross-family pool coverage.

TABLE II
REGISTERED HYPOTHESES AND CONFIRMATORY OUTCOMES.

Hypothesis	Prediction	Outcome
H-specialization *(primary)*	Specialists beat compute-matched generalists	**Null** (p=0.73)
H-compute	3 samples beat 1	Supported on recall (0.63 vs. 0.50), not F1
H-communication	Consensus beats Hierarchical	Unsupported (Holm p=0.12)
H-efficiency	Cost rises along the ladder	Supported (monotonic)
H-complexity	Effect grows with PR complexity	Registered test unavailable; not supported under an exploratory proxy
H-verify	Self-consistency restores parity	**Fails**
H-hetero-prec.	Cross-family agreement lifts precision	**Confirmed** (both sub-claims)

C. Hypotheses and analysis

The design, hypotheses, sample sizes, and analysis plan were pre-registered on OSF and prompts frozen before confirmatory collection. The registered contrasts are summarized with their outcomes in Table II. Two secondary hypotheses were added by a pre-data amendment: **H-verify** (a self-consistency filter lifts each multi-agent arm's F1 to within 0.02 of Agentless) and **H-hetero-precision** (findings corroborated by $\geq$2 of three independent families {Haiku 4.5, Kimi K2.5, GLM 5} gain $\geq$10 precision points at equal-or-higher F1, and all-three agreement exceeds the frozen model's own three-run recurrence by $\geq$10 points). A further pre-data amendment registered a capability contrast (Section IV-E).

The PR is the unit of analysis; contrasts use paired Wilcoxon signed-rank tests with bootstrap CIs and Holm–Bonferroni correction within each hypothesis family, and Cliff's δ as an unpaired magnitude. The registered paired δ differed from the implemented unpaired form; both indicate negligible effects (r=-0.05 vs. δ=-0.005) and do not alter conclusions. All statistics replay deterministically from persisted artifacts with zero new LLM calls.

IV. Results

A. Generation axis: more agents buy recall, never quality

Table III gives per-arm means over the 99 Qodo PRs together with the cost of obtaining them. The **primary hypothesis is null**: at matched compute, Hierarchical ties Generalists-3 on recall (0.624 vs. 0.626; p=0.73, δ=-0.005)

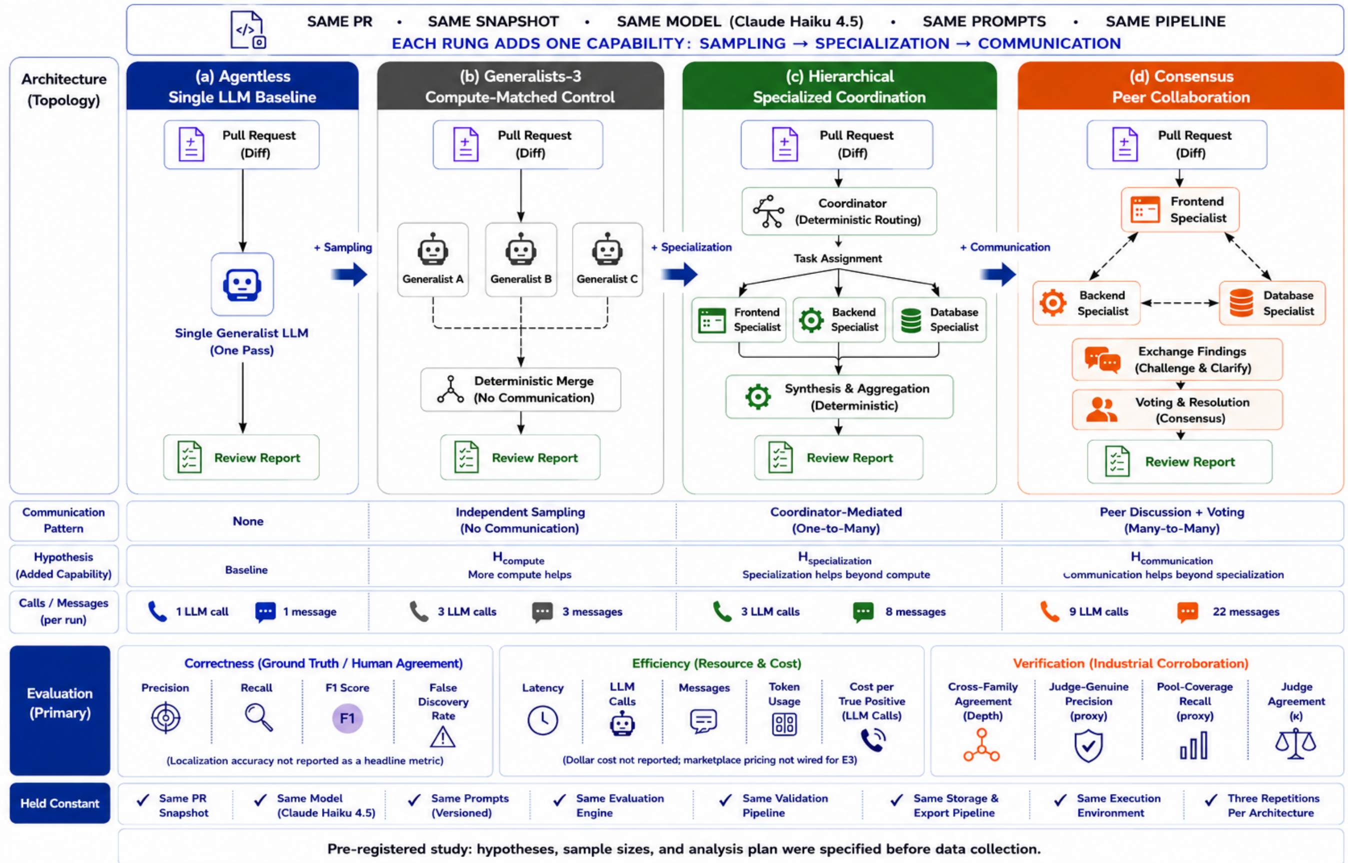

Fig. 1. The four-rung ladder. Adjacent contrasts add sampling, then specialization, then communication, while the PR snapshot, model, prompt templates, and evaluation pipeline are held constant and only the inference budget varies. Coordination and synthesis in the Hierarchical arm are *deterministic*, so its three LLM calls are exactly its three specialists; Consensus adds peer discussion and voting (nine calls). Per-run call and message counts (below each rung) are the overhead in Table III.

at comparable precision. The recall difference is statistically *equivalent* within the pre-registered $\pm$6-point SESOI (TOST $p<0.001$; 90% CI $[-0.025, 0.019]$), so any specialization benefit is below practical significance, not merely undetected—consistent with concurrent evidence that teams average rather than exploit expertise [7]. Nor does compute convert into quality: Agentless dominates every multi-agent arm on semantic F1 (0.487 vs. 0.357–0.378; all Holm $p<0.001$, δ 0.30–0.36). What extra compute buys is recall—Generalists-3 and Hierarchical reach 0.62–0.63 vs. 0.503 (Holm $p<0.001$)—though Consensus, the most expensive arm, does not (0.536, Holm p=0.12): the full peer-consensus workflow did not outperform the hierarchical one despite nine calls vs. three, leaving H-communication unsupported. The trade is stark (Fig. 3a): one confirmed finding costs 0.34 LLM calls under Agentless and 1.76 under Consensus, a 5$\times$ premium for a lower-F1 review; the operational ladder rises monotonically in messages, latency, and cost-per-TP (H-efficiency).

On SWE-PRBench the shape replicates in natural data: Generalists-3 raises coverage of human comments (0.68 vs. 0.51, Holm p=0.001; Hierarchical 0.63, p=0.05), but F1 is a four-way tie (0.325 vs. 0.329–0.357, all n.s., $|\delta|\leq$0.06). The Sonnet 4.5 robustness arm—declared in the pre-registration as a robustness check rather than as a registered hypothesis—has now run at full scale on the same 99 Qodo PRs, and confirms the ordering is not a Haiku artifact: Agentless keeps its F1 dominance (0.546 vs. 0.388–0.425, Holm $p<0.001$, δ 0.32–0.45), the recall-not-F1 pattern holds, and H-verify still fails. H2 fails *harder* rather than identically: the paired median is again 0.000, but Hierarchical now trails Generalists-3 on recall significantly (0.636 vs. 0.672, p=0.012, δ=$-$0.078, CI $[-0.143, -0.017]$)—under the stronger model specialization is not merely no better than compute-matched sampling but measurably worse. Absolute F1 rises (Agentless 0.487$\rightarrow$0.546) and the headline ordering is unchanged, but the ordering *within* the multi-agent arms is not: Consensus and Hierarchical are indistinguishable and change places (0.425 vs. 0.423 under Sonnet; 0.369 vs. 0.378 under Haiku). We therefore claim model-robustness for the registered conclusions and the headline ordering, not for the ordering among the multi-agent arms.

Finally, the pre-registered rescue fails. A self-consistency filter (keep findings recurring in $\geq$2 of 3 runs) leaves Hierarchical at 0.378 ($\tilde{\Delta}$=$-$0.127, CI $[-0.179, -0.083]$) and

TABLE III
E1 (QODO, 99 PRS): REVIEW QUALITY AND INFERENCE OVERHEAD. AGENTLESS IS PRECISION- AND F1-DOMINANT; EXTRA AGENTS BUY RECALL AT 3–9× THE CALLS. FDR = 1−PRECISION; COST/TP = LLM CALLS PER CONFIRMED TRUE POSITIVE. P, R, AND F1 ARE MACRO MEANS OVER PRS (EACH PR THE MEAN OF ITS THREE RUNS), SO F1 IS THE MEAN OF PER-PR F1 AND IS *not* RECOVERABLE AS THE HARMONIC MEAN OF THE P AND R COLUMNS—THE TWO DIFFER BY 2–3 POINTS HERE, AS EXPECTED WHEN F1 IS AVERAGED RATHER THAN DERIVED.

Arm	P	R	F1	FDR	Calls	Msgs	Cost/TP
Agentless	**0.525**	0.503	**0.487**	**0.475**	1	1	**0.34**
Generalists-3	0.270	**0.626**	0.357	0.730	3	3	0.45
Hierarchical	0.294	0.624	0.378	0.706	3	8	0.50
Consensus	0.322	0.536	0.369	0.678	9	22	1.76

Consensus at 0.373 ($\tilde{\Delta}$=−0.114), both entirely below the registered parity band; k=3 fails for every arm. A model that agrees with itself is mostly repeating its confident errors—which motivates the next section.

B. Verification axis: cross-family agreement is a precision instrument

The registered companion generated agentless reviews with two independent frontier families (Kimi K2.5, GLM 5), matched cross-model by a fourth family (Nova Pro) as pair judge to keep the chain non-circular, and tested on the registered 80-PR remainder. **Both sub-claims hold.** Findings corroborated by $\geq$2 families reach per-PR precision 0.716 vs. 0.575 single-arm (n=79 PRs with findings; $\bar{\Delta}$=+0.141, CI $[0.094, 0.189]$, p<10^{-4}, δ=0.364) at equal-or-higher F1, and findings all three families produce are golden-matched **89%** of the time versus **54%** for one model's three-run recurrence (gap +0.348, CI $[0.286, 0.412]$).

Fig. 2 makes the mechanism visible. Cross-family agreement climbs steeply with depth; same-model recurrence stays flat before saturating at its error ceiling. A second run of the same model contributes almost no independent evidence; a second independent family does. The instrument survives a change of pair judge—re-judging all 8,061 candidate pairs with a second pair judge agrees at κ=0.952 and reproduces the rows.

C. Where the signal lives: functional bugs, not conventions

Cross-family agreement concentrates on one class of defect. On the 80-PR set stratified by Qodo's two injected kinds (Fig. 3b), universal *functional* bugs are recovered far more often at every depth than project-specific *rule* violations. The $\geq$2-family set is 52% functional true positives, 20% rule true positives, and 28% golden-unmatched— a high-precision filter for correctness defects, not a substitute for project-specific linting.

D. The recall axis: a decorrelation ladder with a ceiling (exploratory)

Communication settled, we ask how far decorrelated *sampling* alone can reach, and what the residue requires. Unioning K temperature or cross-family draws saturates by three to four

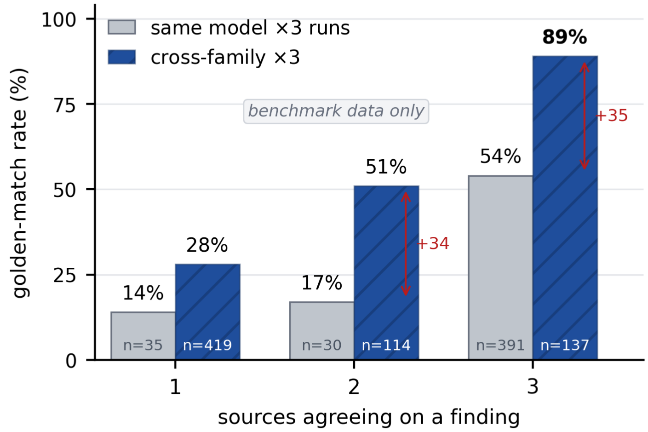

Fig. 2. Benchmark data only (injected-defect ground truth). Agreement predicts truth only when the agreeing sources are *independent*. Repeating one model three times barely improves on a single run until full recurrence, and saturates at that model's error ceiling (54%); three independent families reach 89%. The gap at equal depth—+34 and +35 points—is the correlated-error mechanism made visible. This benchmark relationship reaches its ecological boundary on unlabeled industrial code, where correctness labels are absent (Section IV-H).

sources (Table IV), and the union of *all* eleven configurations we ran (seven model families, a conditioned resampler, a temperature sweep) reaches only 0.83 recall (97 PRs), no single source worth more than ∼2 points by leave-one-out—a robust ceiling. The two numbers measure different axes: Table IV adds one draw per *family* and plateaus at 0.66, whereas 0.83 unions every draw, so the remainder comes from resampling families already in the ladder (the temperature sweep and the conditioned arm), not from new families—the plateau is in the family dimension, and the total-draw ceiling sits above it. (Nova Pro is both a union member and the clustering judge here; this mild circularity is confined to the exploratory union, not the confirmatory contrast of Section IV-B.) The residue is structured: ∼65% lint-targetable conventions, ∼20% conceptual conventions (policy/framework rules), and ∼15% a repo-integrated functional hard-core—36% of rule violations are reached by *no* source, a *correlated* blind spot. A deterministic linter out-recalls the eleven-source union on its target convention rules (0.54 vs. 0.46; hybrid 0.61), so the remaining limitation is evidence, not model diversity—and no precision back-end (agreement or aspect verification) lifts F1 above a plain union. The saturation-to-a-ceiling phenomenon is not Qodo-specific. Replicating the hetero ladder on 50 c-CRAB PRs [15]—a code-review agent benchmark scored here against its human review comments, a different oracle from Qodo's injected defects—reproduces the same shape at a different level (Table IV, bottom row): coverage jumps 16 points on the first added family, plateaus by the second-to-third, and then drifts only from 0.53 to 0.55 across the remaining four. Different oracle, different absolute ceiling, identical plateau.

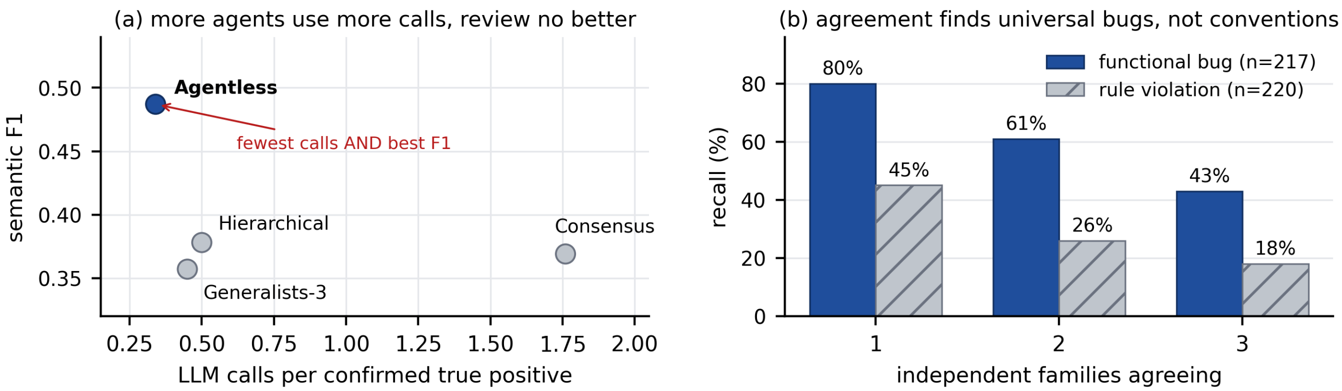

Fig. 3. **(a)** Accuracy/cost frontier: Agentless Pareto-dominates— cheapest (0.34 LLM calls per confirmed true positive) and most accurate (semantic F1 0.487), while Consensus costs 5× more for a worse review. **(b)** Cross-family agreement recovers universal functional bugs far more often at every depth than project-specific rule violations, which a reviewer without repository context cannot know.

TABLE IV

UNION RECALL BY NUMBER OF DRAWS K (SEMANTIC MATCHING). HOMO = ONE MODEL (HAIKU) AT T=0.7; HETERO = HAIKU PLUS ONE DISTINCT FAMILY PER RUNG (KIMI, GLM, DEEPSEEK, NOVA, LLAMA 4, PALMYRA). THE BOTTOM ROW REPEATS THE HETERO LADDER ON A SECOND BENCHMARK WITH A DIFFERENT ORACLE.

K	1	2	3	4	5	6	7
Qodo, 97 PRs (injected defects)							
Homo (T=0.7)	0.49	0.58	0.61	—	—	—	—
Hetero (families)	0.51	0.61	0.64	0.65	0.65	0.66	0.66
c-CRAB, 50 PRs (human review comments)							
Hetero (families)	0.36	0.52	0.53	0.53	0.53	0.55	0.55

The two Qodo ladders start from different single draws: homo K=1 is one Haiku draw at T=0.7, hetero K=1 the T=0 confirmatory anchor—hence 0.49 vs. 0.51. The raw hetero−homo gap is $+0.03$ at K=2, 3, but $+0.02$ of it is that anchor offset; net of it the diversity-attributable increment is $\approx$+0.01 (K=1→3 gains of +0.13 hetero vs. +0.12 homo)—smaller than the raw gap suggests, and consistent with the family-dimension plateau. Family order affects per-step marginals, not the $K=\max$ total. The c-CRAB ladder plateaus by $\sim$3 families, the Qodo hetero ladder by $\sim$4; c-CRAB's level is lower because coverage is scored against sparse human comments.

E. Neither capability nor generation-side structure closes the gap

Three further generation-side interventions also fail to overcome the ceiling. *Capability (registered):* a stronger model does not target functional bugs—on the 80-PR remainder the difference-in-differences is null (DiD=-1.8, CI $[-8.6, 5.5]$, p=0.68), Sonnet 4.5 lifting both types over Haiku equally (functional $0.70 \rightarrow 0.74$, rule $0.32 \rightarrow 0.37$). *Module decomposition (exploratory):* per-module agents unioned beat one whole-PR review by $+6.4$ recall (p=0.016), but only through the $\sim$2.6× compute—at a matched budget module-union *trails* plain multi-sampling (0.54 vs. 0.57, n.s.). *Prompt specialization (exploratory):* convention-only and functional-only personas suppress output (5.3 vs. 16.1 findings/PR) and *reduce* per-type recall (conventions 24% vs. 32%, functional 59% vs. 68%), the team reaching recall 0.47, below the generalist (0.50). Prompt personas are a *negative* decorrelation source: specialization pays only when it changes the detection *mechanism*—a checker, an independent family, or execution—not the prompt.

F. Execution begins to reach what reading cannot (exploratory)

The functional hard-core resists reading in every form we tried. Passive context (whole changed files plus one-hop dependencies) does not raise recall over the diff alone (0.37→0.35, n.s., 50 PRs [15]), and a read-only repository-navigating agent (158 PRs) raises precision (0.17 vs. 0.13) at *lower* recall—agency verifies and prunes, not a coverage lever. Asked to write self-contained reproductions for functional findings, a model succeeds on only $\sim$15% (n=20), declining the rest as inseparable from repository or runtime—the hard-core is genuinely repo-integrated. Real execution changes this: on runtime-error SWE-bench instances whose issue text does not localize the defect, feeding the reviewer the failing-test traceback lifts file-localization recall from 1/6 to 5/6 (25% to 100% where the traceback names the fix file), while on feature-addition instances it is a ceiling (both 100%). Execution recovers exactly the repo-integrated functional defects that diff reading and read-only navigation cannot (small n, single model; directional) [16].

G. How complete is the oracle?

Every precision figure above is measured against the injected golden set, so its completeness bounds their interpretation. We audited it confirmatorily: three independent judge families read every golden-*unmatched* finding against the diff ("does this identify a genuine problem?"), plus a calibration pass on golden-*matched* findings. Llama 3.3 calls 76% of unmatched findings real (91% on known defects); Mistral Large 3 calls 73% (89%); DeepSeek V3.2 calls only 33% (71%).

The disagreement is definitional, not capability: DeepSeek is a strict under-caller (rejecting $\sim$29% of *known* defects) while the lenient judges accept warranted concerns. Inter-judge agreement is poor (κ=0.23–0.58, and under prevalence-robust AC1/PABAK) against the matcher's κ=0.95: whether two findings describe the same issue is judge-stable, whereas whether a finding is a real bug is not settled by an LLM judge of any size.

Three conclusions survive every definition: the golden set is materially incomplete, so golden-referenced precision *understates* all arms equally (relative comparisons intact); the $\geq$2-family set's effective precision is 86–95%; and false positives are less false than they appear (21–32% are localization near-misses, and high-severity findings are almost always real). The magnitude awaits human adjudication; the direction does not—and removing the golden set, as E3 does, leaves only the correctness signal that was already unreliable.

H. Industrial validation: the ecological boundary of cross-family verification

E3 removes the oracle entirely: it asks whether the confirmed cross-family verification signal (Section IV-B) survives when authoritative correctness labels disappear—a boundary test for one relationship, not another dataset comparison. On 30 real merged PRs (627 reviews: the four-arm ladder plus the cross-family axis), scored with the same instruments but proxies for golden labels, two patterns transfer: the communication-cost ladder is unchanged (Agentless 1 call / 13 s to Consensus 9 calls / 173 s), and pool coverage gives the same verdict—more agents do not buy coverage (single-pass highest at 0.27, Consensus lowest at 0.10; pool coverage differs from golden recall, so we compare directions, not ranks).

The cross-family verification signal, however, does not *transfer* to this setting: with no correctness labels the gradient of Fig. 2 cannot be measured. Cross-family agreement is rare: only 23 findings reach depth-2 and 5 reach depth-3 of 904 clusters, versus 114 and 137 on E1. More decisively, the genuineness judges disagree almost completely—Nova labels 99% of findings genuine, DeepSeek 68%, at κ=0.03—so the depth$\rightarrow$precision relation is set by judge choice, not corroboration: flat at 100% under Nova, and *inverted* under the discriminating judge, where findings families agree on score 0% versus 15% for solo findings. This is precisely the boundary Section IV-G anticipated, now with no golden set left to bound it: the objective matcher still works, and the subjective correctness judge does not. The result is therefore about the *measurement apparatus*, not the underlying benchmark relationship, which remains untested rather than overturned—with mutually inconsistent oracles and depth counts of 23 and 5, cross-family agreement can here be neither confirmed nor refuted.

I. Complexity interaction: the effect does not grow with PR complexity

The registered complexity stratifier (PR type) was not persisted, so we proxy it by the number of files an instance's defects span (**exploratory**). No interaction survives on any contrast (all Holm $p\geq 0.42$, $|\delta|\leq 0.12$): complexity degrades every arm by a similar margin and the ordering is stable, so no topology rescues complex PRs. The only directional hint runs *against* the registered prediction (Consensus$-$Hierarchical recall 0.000 on simple vs. -0.119 on complex PRs, Holm p=1.0).

V. Threats to Validity

- **Construct.** E3 is proxy-scored (no oracle), with sparse cross-family agreement (23/5 clusters at depth 2–3), genuineness judges that disagree (κ=0.03), and one partly AI-authored codebase—the conditions it exists to expose. Even the benchmark golden set is incomplete, so golden-referenced precision is a lower bound applying to every arm identically; the pair judge's human validation is pending.
- **Internal.** Model, prompts, evaluation pipeline, and a byte-identical snapshot are fixed while only sampling budget, roles, and communication vary; the compute-matched Generalists-3 control separates specialization from sampling, and evaluation replays deterministically.
- **External.** Qodo is synthetic-injection, so absolute scores sit on a favorable setting (within-benchmark comparisons hold); SWE-PRBench and E3 partly offset this. The Sonnet 4.5 robustness arm has now run at full scale (N=99) and reproduces every registered conclusion (Section IV-A); generalizability of the ordering *within* the multi-agent arms remains a limitation rather than a claim, since Consensus and Hierarchical swap between the two models. The exploratory extensions are small-n, single-benchmark probes (execution directional, n=6). Headline claims are confirmatory (registered sizes, paired tests, CIs, Holm); finer stratifications, RQ3, and the ceiling/capability/execution extensions are exploratory, labeled throughout.

VI. Conclusion

Under a pre-registered, frozen-model design, none of the tested multi-agent workflows improved automated code review. Role specialization adds nothing over compute-matched sampling (the primary null), no arm exceeds the single-pass baseline's F1, extra agents buy recall at a 3–9$\times$ larger per-run call budget, the self-consistency rescue our own pilot suggested does not replicate, and the effect does not grow with PR complexity. The *verification* axis is where multi-agent value lives: agreement between independent model families is a registered precision instrument that survives a change of pair judge—findings all three families produce match injected ground truth 89% of the time versus 54% for one model repeating itself—and the signal concentrates on functional bugs.

That instrument has a precondition. On real unlabeled code it could not be validated—LLM correctness judges disagree at κ=0.03 while the objective matcher stays reliable at κ=0.95—so the industrial result bounds *oracle-based validation*, not cross-family agreement itself. The guidance is therefore conditional: on benchmark-like tasks, do not add homogeneous agents for quality, and treat cross-family agreement as a trust signal to calibrate against a project's own oracle.

The broader lesson is that increasing communication among LLM agents yields diminishing returns once generation saturates; the remaining gains come from expanding what reviewers can *observe*—repository context, executable evidence, and deterministic analyses—rather than from more deliberation among language models. Future work is a fixed-definition *human* adjudication (the oracle these judges could not provide) and scaling the execution probe to confirmatory strength.

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